APARNA RAJ

B.Tech. - Electronics and Telecommunication Engineering

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Pune, Maharashtra, India - 412105

EDUCATION

MIT Academy of Engineering, Pune 2022 - 2026

B.Tech. - Electronics and Telecommunication Engineering | CGPA: **8.04** / **10**

Rajkiyakrit sr. sec. school Gona,Bikram,Patna 2022

12th | Bihar Board | Percentage: **77.60** / **100**

High School Gona,Patna 2020

10th | Bihar Board | Percentage: **84.60** / **100**

INTERNSHIPS

Eduskills | Academics / Research 08 Jan, 2025 - 15 Mar, 2025

Embedded System Developer Intern

Completed an internship focused on designing and developing embedded systems using microcontrollers, C/C++ programming, and hardware interfacing. Gained hands-on experience with real-time systems, sensors, and communication protocols like UART, SPI, and I2C.

Eduskills 05 Oct, 2024 - 26 Dec, 2024

Designed and developed dynamic web applications using Python for backend development and modern frontend frameworks for user interfaces.

Built RESTful APIs to integrate the frontend and backend systems, ensuring seamless data flow and responsiveness.

Implemented database solutions using MySQL, PostgreSQL, MongoDB optimizing data queries and storage for performance and scalability.

Collaborated with cross-functional teams to design, develop, and deploy scalable applications within an Agile environment.

Eduskills | Academics / Research 04 Jul, 2024 - 22 Sep, 2024

Explored and implemented machine learning models, including supervised and unsupervised learning algorithms, to analyze and derive insights from complex datasets.

Developed predictive models using Python libraries such as TensorFlow, Keras, Scikit-learn, and Pandas to solve real-world problems.

Designed, trained, and validated AI models, optimizing performance through hyperparameter tuning and feature engineering.

Worked on projects involving natural language processing (NLP), computer vision, or data analytics, showcasing proficiency in relevant AI/ML techniques.

IIDE (Indian Institute of Digital Education) | Marketing / Lead Management / Data Solutions 15 Jun, 2024 - 15 Jul, 2024

Virtual Intern

The IIDE (Indian Institute of Digital Education) Virtual Internship is a comprehensive, hands-on learning experience designed to introduce students to the essentials of digital marketing and equip them with practical skills. Through this internship, participants gain real-world insights into areas such as Social Media Marketing, Search Engine Optimization (SEO), Content Writing, Data Research, Prospecting, and Google Workspace Management.

PROJECTS

Vehicle Number plate detection using raspberry pi

Mentor: Mr Amit V.M | **Team Size:** 3

Key Skills:

- Currently studying Linux operating systems, focusing on system administration, shell scripting, and process management
- Gaining hands-on experience in command-line interfaces, file systems, and networking.
- Developing a strong foundation in open-source operating systems for future projects.

Remote-Controlled Car Design – Arduino Microcontroller Project

Mentor: Dr Nutan Bhanshode | **Team Size:** 3

Key Skills:

- Designed and built a remote-controlled car using an Arduino microcontroller, integrating sensors and actuators.
- Developed programming skills for motor control, communication, and signal processing.
- Enhanced understanding of robotics, automation, and microcontroller-based systems.

Antenna Design for Breast Cancer Detection – Electromagnetics Project

Mentor: Mr Vipin Yadav | **Team Size:** 4

Key Skills:

- Designed and simulated an antenna using HFSS software for breast cancer detection applications
- Focused on optimizing antenna parameters for high sensitivity and accurate detection.
- Gained experience in electromagnetic wave propagation and simulation tools in medical applications.

Robotic Grass Cutter – Minor Engineering Project

Mentor: Dr Ashish Muljakar | **Team Size:** 4

Key Skills:

- Designed and developed a robotic grass cutter with automated cutting capabilities.
- Implemented sensors and motor control systems for efficient and precise grass cutting.
- Enhanced skills in robotics, automation, and system integration for real-world applications.

AI-Powered Law Chatbot-Major Project

Mentor: Dr N.D Kale | **Team Size:** 3

Key Skills:

Machine Learning | Chatbot Development | Legal Domain Knowledge | Python Programming | Testing and Debugging | Web Scraping

Developed an AI chatbot using NLP to answer legal queries, classify topics, and provide legal guidance. Built with Python, Javascript, and Api environment for web deployment.

ASSESSMENTS / CERTIFICATIONS

Web Devlopment

MATLAB Onramp

Internship & Job Preparation

Programming Essential in Python

Java Basic Program

Introduction to Cybersecurity

Introduction to Internet of Things

Aggregate: 75 / 100

Introduction to Microsoft Excel

Certificate of Participation

Certificate of Completion

Course Completion Certificate

IMs

- Other - https://www.linkedin.com/in/aparna-raj-a13613258/

PERSONAL DETAILS

Gender: Female
Marital Status: Single
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